Project Proposal – Bike Rentals Predictor (Daily Forecast)

Table of Contents

Introduction

What?

I am going to build an AI project called **Bike Rentals Predictor**, which forecasts the number of bike rentals at a given day based on measurable **weather** and **time-related features**. The project will use a historical dataset to train a model that can highlight the relationship between factors like temperature, humidity, season, and demand for bikes.

Why?

Bike-sharing companies and city planners face the challenge of managing supply and demand. Too few bikes at stations cause frustration, while too many lead to wasted resources. By creating a data-driven model, this project will provide **concrete predictions** of bike rental demand, making it easier to:

- Optimize bike distribution across stations.
- Improve urban mobility planning.
- Ensure citizens have better access to bikes at peak times.

Who is involved?

- **Bike-sharing companies** → to plan supply and avoid shortages.
- **City transport planners** → to optimize urban infrastructure.
- **Fatih Koran (Koran, Fatih Koran F.)** → project owner and researcher.

How?

Project Approach

The project will focus on developing an AI tool that predicts daily bike rentals using historical weather and time data.

How it will work and what I will do:

- Collect and analyze the **UCI Bike Sharing Dataset** (Washington D.C., 2 years).

- Clean and prepare the data, including features such as temperature, humidity, windspeed, season, month, weekday, and hour.
- Train a supervised machine learning model to predict the target variable: **cnt (total number of rentals)**.
- Test model performance against unseen data and validate predictions with logical checks.
- Evaluate performance based on **accuracy, usability, and relevance**.
- Develop a prototype (dashboard or notebook) that provides **predictions and insights**.
- Present results and refine based on feedback.

Project Management

- Define iteration goals (proposal, data preparation, modeling, delivery).
- Break tasks into smaller steps and track them with **Trello**.
- Reflect after each iteration to identify improvements.
- Document progress to ensure **transparency and reproducibility**.

When?

The project runs until Week 9, the final week for presenting results. I am currently in Week 4. The timeline is structured as follows:

- Week 4 → Finalize proposal and secure dataset (or start survey if needed).
- Week 5–6 → Data provisioning and cleaning (collect, preprocess, and explore data).
- Week 6–7 → First modeling attempts (Linear Regression and baseline evaluation).
- Week 7–8 → Iterative improvements (feature engineering, testing additional models, refining evaluation).
- Week 8–9 → Prototype development (dashboard/notebook), validation, and preparation of presentation.
- Week 9 → Final presentation and delivery.

Throughout the project, I will not work strictly step-by-step. Instead, I will follow an iterative approach: every two weeks I will update progress across all project stages (data, modeling, evaluation, and reporting). This ensures continuous improvement and avoids last-minute bottlenecks.

Domain Understanding

Focus Points

- Weather and time-related factors are the key drivers of daily rental demand.
- Providing accurate daily forecasts helps optimize bike distribution and reduce shortages or excess supply.
- The insights can guide city planners in sustainable transportation strategies.

Methodologies

- Literature study on demand forecasting in bike-sharing systems.
- Data collection and preparation from the UCI Bike Sharing Dataset.
- Iterative modeling process (starting with Linear Regression, with the option to extend to other algorithms).
- Regular evaluation and refinement using error metrics.
- Feedback incorporation after prototype testing.

Summary

Traditional planning of bike distribution is often inefficient and reactive. By applying AI to predict daily rental demand based on weather and time features, the process becomes data-driven, faster, and more reliable. This ensures better service for citizens, improved urban mobility, and more efficient use of resources.

Analytic Approach

Modeling Type:

- This is a **supervised regression** problem, since the goal is to predict a numeric count of rentals.

Target Variable:

- cnt → total number of **daily rentals**.

Input Features:

- **Weather factors:** temperature, humidity, windspeed
- **Time factors:** season, month, weekday, hour
- These capture both environmental and temporal influences on daily demand.

Modeling Approaches:

- Start with **Linear Regression** for interpretability and as a baseline.
- If performance is insufficient, extend to more advanced models such as **Random Forest Regressor** or **Gradient Boosting**, which can capture non-linear relationships.
- Iteratively test and compare models to balance interpretability and accuracy.

Process:

1. Explore dataset with descriptive statistics and visualizations.
2. Preprocess the data: normalization of continuous features, encoding of categorical features, and handling of missing values.
3. Train the model on training data and evaluate it on test data.
4. Compare models and refine based on performance.

Evaluation Metrics:

- **RMSE (Root Mean Squared Error):** Measures average prediction error size. Lower = better.
- **R^2 (Coefficient of Determination):** Explains how much variance in rentals is captured by the model. Higher = better.
- **MAE (Mean Absolute Error):** Provides a straightforward measure of average daily prediction error

Reasoning:

- Linear Regression clearly shows how much each factor (temperature, season, weekday, etc.) contributes to **daily demand**. If needed, more complex models can be added to capture hidden interactions.

Outcome:

- The analysis will generate **daily forecasts of bike rentals** and highlight which weather and time-related factors have the greatest influence on demand.

Data Requirements

- **Target Variable**: cnt → total number of rentals.
- **Input Features**:
    - temp (temperature in °C)
    - humidity (%)
    - windspeed (m/s)
    - season (1–4)
    - month (1–12)
    - weekday (0–6)
    - hour (0–23)
- **Source**: UCI Bike Sharing Dataset (17,000+ hourly rows, Washington D.C., 2011–2012).
- **Data Quality Needs**:
    - No missing values allowed.
    - Continuous variables normalized.
    - Categorical variables properly encoded.
    - Aggregation from hourly data into daily forecasts must be consistent.
- **Ethics**:
    - The dataset is public and anonymized, containing no personal or sensitive data, which ensures the project respects privacy and responsible AI use.

Deliverables

- **Project Proposal and Strategy Plan** → A detailed document outlining objectives, methodology, and timeline.
- **Cleaned and Documented Dataset** → UCI Bike Sharing data prepared for analysis (daily focus, with all preprocessing steps applied).
- **Exploratory Data Analysis (EDA) Report** → Visualizations and statistical summaries showing how weather and time features affect daily rentals.
- **Trained Regression Model(s)** → Baseline Linear Regression model, plus potential alternative models (Random Forest, Gradient Boosting) for comparison.
- **Evaluation Report** → Model results with RMSE, R^2, and MAE, including interpretation of which features most strongly influence demand.
- **Prototype Dashboard/Notebook** → A simple interactive tool or notebook that demonstrates daily predictions of bike rentals.
- **Final Presentation and Demo** → Summary of approach, results, and prototype demonstration, presented to stakeholders at the end of the project.